

Doping Stability Report -- Ca in LiCoO₂ at 350 C

Generated: 2026-05-30 22:09 Host: LiCoO₂ Dopant: Ca

Executive Summary

Ca preferentially occupies the Co layer. The formation energy difference between the two layers is +3.432 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00372 Å²/ps (stable), compared to -0.00206 Å²/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-1.761	-1	No	PASS	FAIL	STRUCTURAL COLLAPSE
Li	+1.671	+1	Yes	PASS	PASS	METASTABLE

Ca at the Co layer -- STRUCTURAL COLLAPSE

Charge situation

Ca replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E_f	-1.7611 eV	PASS	< 1.0 eV to pass
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E_f = -1.761 eV. This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	-0.00372 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.568 Å²	----	
Max displacement	1.656 Å	----	
Coordination (min/mean/max)	5 / 6.1 / 8	FAIL	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.148 Å (relaxed: 2.1)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00372 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.568 Å²; maximum displacement from starting position = 1.66 Å.

Coordination fluctuated between 5 and 8 (mean 6.1), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.148 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

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The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ca atom barely moves (MSD = 0.568 Å²) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

? **WARNING:** Charge deficit (-1) cannot be compensated in CHGNet (would require adding Li interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Ca at the Li layer -- METASTABLE

Charge situation

Ca replaces Li with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E _f	+1.6712 eV	FAIL	< 1.0 eV to pass
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E_f = +1.671 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Ca incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	-0.00206 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.089 Å²	----	
Max displacement	0.752 Å	----	
Coordination (min/mean/max)	5 / 6.0 / 7	PASS	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.220 Å (relaxed: 2.2)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00206 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.089 Å²; maximum displacement from starting position = 0.75 Å.

Coordination fluctuated between 5 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.220 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: METASTABLE

Ca sits stably on the Li site during MD at 350 C, but the formation energy exceeds the 1.0 eV equilibrium threshold. This configuration is metastable -- it may be achievable via non-equilibrium synthesis routes such as ion exchange or rapid quenching.

Site Preference Comparison

The Co site is preferred by 3.432 eV over the Li site. Formation energy: -1.761 eV (Co) vs +1.671 eV (Li).

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approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.